

# POINTER.md — novus-aesc Subsystem Specification

Repository: novus-aesc (APK 2 — Native System Terminal & Shell Daemon)

Authority: 01\_SOVEREIGN\_NODE\_AND\_APK\_TOPOLOGY.md &

05\_FEDERATED\_FILE\_TREE\_TOPOLOGY\_MASTER.md

## ## Core Architectural Invariants

1. ForegroundService Execution: Runs with START\_STICKY and FOREGROUND\_APP\_ADJ to ensure 100% resilience against Android Low Memory Killer (LMK).
2. ADB Loopback (127.0.0.1:5555): Maintains persistent loopback connection to execute shell commands with UID 2000 permissions without requiring root.
3. NPU Watchdog: Operates npu\_manager.py over UNIX domain sockets (/dev/socket/npu\_manager.sock) for hot-swapping open-weights on Qualcomm Hexagon NPU.
4. Laptop Trick Tunnel: Routes reverse-proxy shell connections and debugging telemetry across local network interfaces.
5. Ingestion & Normalization: Awaits Phase 1 extraction and script migration from Termux environment.